COMSATS University, Islamabad Pakistan

OVULITE

By

Eman Malik **CIIT/SP23-BDS-012/ISB**

Inshal Zafar **CIIT/SP23-BDS-021/ISB**

Supervisor

Dr. Nusrat Shaheen

Bachelor of Science in Data Science(2023-2027)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.

COMSATS University, Islamabad Pakistan

OVULITE

A project presented to

COMSATS University, Islamabad

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Data Science(2023-2027)

By

Eman Malik **CIIT/SP23-BDS-012/ISB**

Inshal Zafar **CIIT/SP23-BDS-021/ISB**

DECLARATION

We hereby declare that this software, neither whole nor as a part, has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be a reproduction of some other, we will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Eman Malik

Inshal Zafar

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) Dukan was developed by **Eman Malik (CIIT/SP23-BDS-012)** and **Inshal Zafar (CIIT/SP23-BDS-021)** under the supervision of Dr. Nusrat Shaheen and that in (their/his/her) opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences / Software Engineering.

Dr. Nusrat Shaheen

Supervisor

External Examiner

Dr Farhana Jabeen

Head of Department

(Department of Computer Science)

EXECUTIVE SUMMARY

The Ovulite platform is a comprehensive, AI-powered reproductive intelligence system designed to modernize the bovine embryo transfer process by providing objective grading, accurate success predictions, and automated quality auditing. The system integrates advanced machine learning architectures, specifically utilizing **PyTorch** for computer vision and **XGBoost** for clinical tabular analysis, to standardize embryo assessment and pregnancy forecasting. By leveraging a primary dataset of standardized clinical records, Ovulite identifies key physiological "sweet spots"—such as optimal Corpus Luteum (CL) size and Progesterone (P4) levels—to assist veterinarians in selecting the most viable recipients for transfer.

The platform is structured into seven core functional modules, including a **Pregnancy Success Prediction Engine**, **AI Embryo Grading via Image-Metadata Fusion**, and an **Unsupervised Lab QC Indicator**. These modules provide veterinarians, lab technicians, and farm managers with real-time, data-driven insights that eliminate human subjectivity and reduce embryo wastage. While the grading engine employs **Deep Learning** to analyze day 6–8 embryos, the Lab QC module utilizes **Isolation Forest** algorithms to detect anomalies in media batches or operator performance, ensuring the highest standards of laboratory precision.

Furthermore, Ovulite includes specialized tools for **Protocol Effectiveness Analysis** and a **Reproductive Intelligence Dashboard**. These features empower users to evaluate synchronization strategies and visualize the entire IVF-to-Pregnancy funnel through high-fidelity, glassmorphic interfaces. The system also features a **Continuous Learning Loop**, ensuring that as final pregnancy outcomes are recorded, the underlying AI models are monitored for performance drift and retrained to maintain peak predictive accuracy.

Developed with a modern, high-performance tech stack, the platform utilizes **FastAPI** and **Python** for a seamless backend and ML integration, while **PostgreSQL** ensures robust management of complex clinical metadata. The frontend is built using **Next.js** and **Tailwind CSS**, providing a responsive and high-tech user experience. For deployment, the entire ecosystem is containerized using **Docker**, ensuring that Ovulite remains a scalable, reproducible, and vital tool for improving the efficiency and success rates of bovine embryo transfer operations globally.

.

Abstract

The rapid expansion of In-Vitro Fertilization (IVF) and Embryo Transfer (ET) in Pakistan's cattle industry has exposed critical challenges in reproductive decision-making, where technicians and embryologists still rely heavily on subjective visual assessment and manual judgment, leading to inconsistent pregnancy outcomes and significant economic losses. The proposed system, Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle, aims to resolve these challenges by transitioning from subjective methods to an objective, data-driven framework. By analyzing a standardized clinical dataset of 488 records across 39 variables, the platform leverages machine learning to provide real-time insights into the IVF-ET cycle through seven integrated modules. These include an AI-based Embryo Grading engine utilizing image-metadata fusion (CNN/ResNet18), a Pregnancy Success Prediction engine (XGBoost/TabPFN) that identifies clinical "sweet spots" in recipient physiology, and an unsupervised Lab QC Indicator employing anomaly detection (Isolation Forest) to flag media or operator-related drifts. Unlike traditional herd management tools, Ovulite focuses specifically on embryo-level intelligence, offering protocol effectiveness analysis and a unified reproductive dashboard to visualize the OPU-to-Pregnancy funnel. Ultimately, this project establishes a Continuous Learning Loop that monitors model performance and facilitates retraining based on final outcomes, reducing human error and supporting Pakistan's transition toward precision livestock reproduction and data-centric farm management.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Dr. Nusrat Shaheen” . Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are grateful to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us with the values of honesty & hard work.

Eman Malik

Inshal Zafar

Abbreviations

SRS	Software Require Specification
PC	Personal Computer
ET	Embryo Transfer
IVF	In Vitro Fertilization
CL	Corpus Luteum
OPU	Ovum Pick-up
CIDR	Controlled Internal Drug Release
PGF2a	Prostaglandin F2 Alpha
GnRH	Gonadotropin-Releasing Hormone
AI	Artificial Insemination
BCS	Body Condition Score
SHAP	SHapley Additive exPlanations
CNN	Convolutional Neural Network
ROC	Receiver Operating Characteristic
API	Application Programming Interface
RAG	Red-Amber-Green
NFR	Non-Functional Requirement
SDLC	Software Development Life Cycle
JWT	JSON Web Token
REST	Representational State Transfer
AWS	Amazon Web Service

Table of Contents

Abstract 5
Chapter 1: Introduction and Problem Definition 4
1.1 Overview of the Project 4
1.2 Vision Statement 4
1.3 Problem Statement 4
1.4 Problem Solution 5
1.5 Objectives of the Proposed System 5
1.6 Scope 5
1.6.1 Limitations 6
1.6.2 Constraints 7
1.7 Modules 7
1.7.1 Module 1: Pregnancy Success Prediction Engine 7
1.7.2 Module 2: Secure Access & Role Management 8
1.7.3 Module 3: AI Embryo Grading (Image + Metadata Fusion) 8
1.7.4 Module 4: Lab QC Indicator 8
1.7.5 Module 5: Protocol Effectiveness Analysis 8
1.7.6 Module 6: Reproductive Intelligence Dashboard 9
1.7.7 Module 7: Outcome Tracking, Traceability & Learning 9
1.8 Related System Analysis/Literature Review 9
1.8.1 Literature Review 9
1.8.2 Related System Analysis 10
1.9 Tools and Technologies 10
1.10 Project Contribution 12
1.11 Relevance to Course Modules 13
Chapter 2: Requirement Analysis 15
2.1 User Classes and Characteristics for Ovulite Project 15
2.1.1 Use case Diagram 16
2.1.1.1 Actor: Veterinarian 16
2.1.1.2 Actor: Farm Manager 17
2.1.1.3 Actor: Embryologist 18
2.1.1.4 Actor: System Adminitrator 19
2.1.2 Requirement Identifying Technique 19
2.1.3 Backend Process Requirements 20
2.2 Functional Requirements 21
2.2.1 Mockups: 22
2.2.1.1 Mockups M1 – Login Page Screen 22
2.2.1.2 Mockup M2 –Dashboard Page 23
2.2.1.3 Mockup M3– ET Transfer Records 25
2.2.1.4 Mockups M4 – Reports & Export 26
2.2.1.5 Mockups M5 – Case Records & Traceability 27
2.2.1.6 Mockup M6- Embryo Grading 28
2.2.1.7 Mockup M7- Lab QC & Anomaly Detection 30

2.2.1.8 Mockup M8 - Data Entry - New ET Transfer 31
2.2.1.9 Mockup M9- Pregnancy Prediction 33
2.2.1.10 Mockup M10-Analytical Dashboard 34
2.2.1.11 Mockups M11 – Analytics Dashboard(Protocol+ Donors + Biomarkers) 35
2.3 Non-Functional Requirements 37
2.3.1 Reliability 37
2.3.2 Usability 38
2.3.3 Performance 38
2.3.4 Security 38
2.3.5 Accessibility 38
2.3.6 Portability 39
2.3.7 Maintainability 39
2.4 External Interface Requirements 39
2.4.1 User Interfaces Requirements 39
2.4.2 Software interfaces 40
2.4.3 Hardware interfaces 40
2.4.4 Communications interfaces 40

List of Figures

Figure 1 Mockup M1 for Login Page 21
Figure 2 Mockup M2 for Dashboard Page 23
Figure 3 Mockup M3 for ET Transfer Records 24
Figure 4 Mockup M4 for Reports & Export 25
Figure 5 Mockup M5 for Case Records & Traceability 26
Figure 6 Mockup M6 for Embryo Grading 27
Figure 7 Mockup M7- - Lab QC & Anomaly Detection 28
Figure 8 Mockup-8 Data Entry - New ET Transfer 29
Figure 9 Mockup-9 Pregnancy Prediction 31
Figure 10 Mockup-10 Analytical Dashboard 3236

List of Tables

Table 1 Related System Analysis with proposed project solution 11
Table 2 Tools and Tecnologies 11
Table 3 Ovulite userclass and Characteristics 15
Table 4 Event Response Table 20
Table 5 Functional Requirements Derive from Mockup M1 22
Table 6 Functional Requirements Derive from Mockup M2 23
Table 7 Functional Requirements Derived from Mockup M3 24
Table 8 Functional Requirements Derived from Mockup M4 25
Table 9 Functional Requirements Derived from Mockup M5 26
Table 10 Functional Requirements Derive from Mockup M6 27
Table 11 Functional Requirements Derived from Mockup M7 28
Table 12 Functional Requirements Derived from Mockup 8 31
Table 13 Functional Requirements Derived from Mockup-9 32
Table 14 Functional Requirements Derived from Mockup-10 33
Table 15 Functional Requirements Derived from Mockup 11 36

Chapter 1: Introduction and Problem Definition

The Ovulite platform is an AI-powered reproductive intelligence system designed to modernize bovine embryo transfers by automating grading and success prediction. Utilizing a 7-module architecture, the system replaces subjective manual judgment with objective insights from **CNN-based embryo grading** and **XGBoost-driven pregnancy forecasting**. By integrating clinical inputs like CL size and progesterone levels, Ovulite provides veterinarians and farm managers with a data-driven framework to reduce embryo wastage and optimize breeding efficiency.

1.1 Overview of the Project

The Ovulite is a specialized management platform built to improve the accuracy of the IVF-ET cycle in the livestock industry. Leveraging a **FastAPI** backend and a high-fidelity **Next.js** interface, the platform serves predictive models that analyze multi-modal data to forecast pregnancy outcomes and detect laboratory anomalies. The project focuses on providing actionable reproductive analytics—including protocol effectiveness and technician performance—through a unified dashboard. By establishing a **Continuous Learning Loop**, Ovulite ensures that AI models evolve with new clinical data, offering a scalable solution for precision livestock reproduction.

1.2 Vision Statement

For veterinarians, farm managers, and lab technicians who need to optimize the accuracy and efficiency of bovine embryo transfer procedures, the Ovulite platform is an AI-powered solution that automates embryo grading, predicts pregnancy outcomes, and streamlines data management. Unlike traditional manual methods, our product leverages machine learning to provide real-time, objective insights, reducing human error and enhancing breeding success rates for livestock farms.

1.3 Problem Statement

The livestock breeding process, especially in the context of bovine embryo transfer, relies heavily on manual data entry, subjective grading of embryos, and limited access to advanced decision support tools. This leads to inefficiencies, increased chances of error, and suboptimal breeding outcomes. Veterinarians and lab technicians often struggle to process large amounts of data in real time, while farm managers lack actionable insights to optimize their herd's reproductive health.

Ovulite solves these problems by automating the grading process using AI, providing real-time pregnancy predictions, and offering a comprehensive dashboard for farm managers. The system reduces human error, streamlines the embryo transfer process, and improves overall farm productivity.

1.4 Problem Solution

The Ovulite system automates critical tasks such as:

- **Embryo Grading:** Using AI to evaluate embryos and determine their quality for transfer.
- **Pregnancy Prediction:** AI-powered predictions on the likelihood of pregnancy based on embryo quality.
- **Lab Quality Auditing:** Leveraging unsupervised anomaly detection to monitor process health, flagging deviations in media batches or operator performance.
- **Improved Decision Support:** Enhanced decision-making through data-driven insights, helping veterinarians, farm managers, and lab technicians make better clinical decisions.

By implementing Ovulite, the system will significantly improve the embryo transfer process, ensuring higher success rates, reducing operational costs, and enhancing the overall efficiency of livestock breeding operations.

1.5 Objectives of the Proposed System

The Ovulite platform aims to achieve the following objectives:

BO-1: Develop a machine-learning model (TabPFN/XGBoost) to predict bovine pregnancy success using standardized clinical inputs like CL size, blood flow, and P4 levels.

BO-2: Implement an AI-driven embryo grading system using **Image-Metadata Fusion** (CNN/ResNet18) to provide objective quality scores (1, 2, or 3).

BO-3: Design a **Lab QC Indicator** using unsupervised anomaly detection (Isolation Forest) to monitor process health and detect media or technician-related shifts.

BO-4: Analyze the effectiveness of different synchronization protocols to support data-driven decisions on hormone treatments and embryo types.

BO-5: Create a unified **Reproductive Intelligence Dashboard** to visualize KPIs, including the IVF-to-Pregnancy funnel and technician performance metrics.

BO-6: Establish a continuous learning loop that utilizes real-world pregnancy outcomes to retrain and improve model accuracy over time.

1.6 Scope

The scope of Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle encompasses the development of a unified decision-support platform that transforms raw clinical records into actionable reproductive insights. The system will process donor, embryo, recipient, technician, and hormonal data already collected by farms and IVF laboratories to deliver a structured and standardized data environment. Key functionalities include machine learning-based pregnancy success prediction using clinical indicators like CL size and

progesterone levels, and automated embryo grading through a computer vision pipeline that utilizes image-metadata fusion. Ovulite will generate reproductive analytics such as protocol effectiveness trends, embryo day/stage success patterns, and technician performance insights to support evidence-based management decisions. The system will visualize "sweet spots" and failure patterns in existing reproductive workflows, helping veterinarians identify the optimal conditions for successful transfers.

Additionally, Ovulite will incorporate a comprehensive reproductive intelligence dashboard for monitoring KPIs, including OPU-to-pregnancy funnel analytics and lab process health. The platform will provide a structured web-based interface for embryologists to enter embryo images and for veterinarians to update recipient profiles and pregnancy outcomes, effectively closing the feedback loop. The solution does not include real-time sensor integration, genomic analysis, abortion risk assessment, sire compatibility matrices, or automated physiological monitoring, as these are beyond the dataset and resources available in the current project. It also excludes financial modules unrelated to reproduction, such as full herd management, milk yield tracking, or feedlot operations. The project will deliver a functional, Docker-containerized prototype containing core predictive models, visual dashboards, and secure user interfaces required for practical use by IVF labs and farm managers. By defining clear functional boundaries, the project ensures feasibility within the available data and academic timeline while offering high-impact, data-driven reproductive intelligence for the cattle industry.

1.6.1 Limitations

LI-1: The Ovulite platform is specifically optimized for bovine embryo transfers; adapting the predictive models for other livestock, such as goats or sheep, would require significant retraining and a different set of physiological parameters.

LI-2: The accuracy of the Pregnancy Success Prediction and AI Grading modules is dependent on the quality of the 488-row historical dataset; any systemic biases or missing values in clinical inputs like Progesterone (P4) levels could affect model reliability.

LI-3: While the system provides high-tech insights, it does not replace clinical ultrasound or professional veterinary judgment, but rather serves as a decision-support tool to complement human expertise.

LI-4: The Lab QC Indicator relies on a consistent stream of outcome data to detect anomalies; in environments with very low case volumes, the unsupervised learning model may take longer to identify statistically significant performance shifts.

LI-5: The current version of the platform does not support real-time integration with external laboratory hardware or automated genetic sequencing tools, requiring manual entry of embryo metadata and images.

LI-6: AI model performance is subject to "drift" over time as farm conditions or cattle breeds change; the system requires periodic retraining through the outcome tracking module to maintain high predictive precision.

1.6.2 Constraints

CO-1: Deployment: Ovulite must be deployed as a containerized solution using Docker to ensure that the complex Python-based AI dependencies (FastAPI, PyTorch, XGBoost) remain stable across different environments.

CO-2: Data Privacy: The platform must adhere to general data privacy standards, ensuring that sensitive farm-level clinical data and technician performance records are protected via Role-Based Access Control (RBAC).

CO-3: Connectivity: As a web-based application (Next.js/FastAPI), the system requires a stable internet connection for users to access the centralized PostgreSQL database and run cloud-based AI inferences.

CO-4: Technical Stack: The system is constrained to the use of Python-native libraries for its backend to ensure seamless integration with the machine learning models and the SHAP-based explainability engine.

CO-5: Data Structure: All incoming clinical data must be mapped to the standardized 39-column schema established during the data audit to ensure the predictive models receive valid inputs.

CO-6: The platform will adhere to general data privacy standards but will not provide specific compliance features for every regional data protection law, requiring users to ensure local legal compliance.

1.7 Modules

The proposed system consists of seven core modules that collectively deliver AI-driven reproductive intelligence for cattle. Each module is designed to transform raw IVF–ET–pregnancy data into meaningful predictions, performance alerts, and clinical analytics. The system is divided into functional components that cover data security, embryo assessment, pregnancy prediction, lab quality control, protocol effectiveness, and outcome learning. The following modules and their features define the precise boundaries of the Ovulite system.

1.7.1 Module 1: Pregnancy Success Prediction Engine

This module predicts the likelihood of pregnancy based on standardized clinical inputs and historical patterns.

- **FE-1:** Predict pregnancy outcome (Pregnant/Open/Recheck) using CL size, blood flow score, P4 level, embryo metadata, technician, and protocol.
- **FE-2:** Display prediction probability as a percentage and a RAG (Red-Amber-Green) confidence score.
- **FE-3:** Provide SHAP-based explanatory factors to visualize which clinical variables (e.g., CL size) most influenced the specific prediction.

- **FE-4:** Calibrate predictions against historical accuracy to provide a "Model Reliability" band for the user.

1.7.2 Module 2: Secure Access & Role Management

This module ensures secure access to the platform, managing user sessions based on assigned clinical roles.

- **FE-1:** Secure user login via encrypted credentials to ensure only authorized personnel can access farm data.
- **FE-2:** Role-Based Access Control (RBAC) to differentiate permissions between Admin (Farm Owners) and Operators (Veterinarians/Technicians).
- **FE-3:** Active session management and secure logout functionality to protect data integrity.

1.7.3 Module 3: AI Embryo Grading (Image + Metadata Fusion)

This module leverages computer vision to automate embryo grading by combining visual features with developmental metadata.

- **FE-1:** Upload and process embryo images (Day 6, 7, or 8).
- **FE-2:** Extract deep-learning features using a Convolutional Neural Network (CNN) such as ResNet18.
- **FE-3:** Combine visual features with metadata (Day, Stage, Media Type) for a holistic viability assessment.
- **FE-4:** Predict objective embryo grades (1, 2, or 3) to eliminate human subjectivity.
- **FE-5:** Generate Grad-CAM heatmaps showing the specific regions of the embryo image that contributed to the assigned grade.

1.7.4 Module 4: Lab QC Indicator

This module uses unsupervised machine learning to monitor lab performance and identify process deviations in real-time.

- **FE-1:** Detect abnormalities in lab performance metrics (e.g., media lot success, technician success rates).
- **FE-2:** Use unsupervised anomaly detection (Isolation Forest) to flag batches that deviate from the historical baseline.
- **FE-3:** Monitor for "Model Drift" to identify when changes in the environment are affecting predictive accuracy.
- **FE-4:** Generate automated quality audit reports for lab managers.

1.7.5 Module 5: Protocol Effectiveness Analysis

This module evaluates the performance of various synchronization and transfer protocols to optimize breeding strategies.

- **FE-1:** Compare success rates between conventional and sexed embryos across different recipient groups.
- **FE-2:** Evaluate synchronization protocol effectiveness (e.g., OPU-Standard vs. Enhanced) on pregnancy outcomes.
- **FE-3:** Analyze the interaction between technician performance and specific protocols.
- **FE-4:** Provide success dashboards showing pregnancy percentages and failure patterns per protocol.

1.7.6 Module 6: Reproductive Intelligence Dashboard

This module provides a unified visual command center for farm managers and veterinarians.

- **FE-1:** Visualize "Sweet Spot" charts for CL-size and P4-levels to identify optimal transfer windows.
- **FE-2:** Display the IVF Funnel (OPU → Embryo → ET → Pregnancy) to identify where the highest losses occur.
- **FE-3:** Track embryo batch-level success and identify operator-specific success trends.
- **FE-4:** Consolidate all reproductive KPIs into a single, high-tech interface for rapid decision-making.

1.7.7 Module 7: Outcome Tracking, Traceability & Learning

This module closes the loop by tracking final outcomes and enabling the system to learn from new data.

- **FE-1:** Generate a unique Case ID for every transfer to ensure 100% traceability.
- **FE-2:** Maintain a complete timeline of events from OPU to final pregnancy confirmation.
- **FE-3:** Record actual pregnancy outcomes to compare against AI predictions.
- **FE-4:** Track model performance metrics (Accuracy, ROC-AUC, F1-Score) as new data is added.
- **FE-5:** Trigger model retraining logs when significant new outcome data is available to improve future accuracy.

1.8 Related System Analysis/Literature Review

1.8.1 Literature Review

Several digital and AI-enabled livestock solutions exist globally, but none provide an integrated embryo-level reproductive intelligence system specifically designed around Pakistan's cattle IVF and ET workflows. **EmGenysis** is an embryo monitoring and management platform offering digital embryo records and donor tracking; however, it lacks AI-based pregnancy prediction, abortion risk alerts, and recipient suitability analysis, making it insufficient for data-driven ET decisions. **Folio3 AI Livestock** Management provides herd monitoring, disease alerts, and vision based animal identification, but does not support IVF-related analytics, embryo grading, or ET success modeling. **EmbryoPlus,** a commercial bovine reproduction service, offers advanced

IVF services and donor management but does not provide farm-level predictive models or AI-driven embryo–recipient matching based on physiological and hormonal data. **Taurus Herd Health** focuses on herd fertility monitoring and insemination tracking, yet it does not incorporate embryo image analysis, donor–sire matrix optimization, or technician performance analytics. These limitations highlight the global gap in integrated, AI-supported embryo-level reproductive intelligence. **Ovulite** addresses this gap by merging embryo-quality assessment, pregnancy prediction, s into one unified reproductive intelligence system powered by machine learning and image fusion models.

1.8.2 Related System Analysis

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Folio3 AI Livestock Management	Focuses on herd health and monitoring, not IVF/ET-specific reproductive analytics or embryo-quality evaluation.	Integrates **Multi-modal Fusion** (image + metadata) for embryo grading and ET-specific predictive success modeling.
EmbryoPlus	Provides IVF services but lacks predictive intelligence, in-lab embryo viability scoring, and recipient suitability modeling.	Adds AI-based embryo grading, **CL/P4-based suitability scoring**, and technician performance analytics.
Taurus Herd Health	Taurus Herd Health Tracks fertility cycles but does not handle embryo images, IVF workflows, ET success prediction, or donor–sire matrix.	Extends functionality with advanced embryo analytics, **ML-driven success prediction**, and a unified reproductive intelligence dashboard.
EmGenysis	EmGenysis Offers embryo records but no AI pregnancy prediction.	Provides **TabPFN/XGBoost-based pregnancy prediction**, abortion risk alerts, and an AI-driven matching engine.

1.9 Tools and Technologies

Table 1 Tools and Tecnologies

	Tools and Technologies	**Version**	**Rationale**
	Frontend Development		
	Next.js	13+ (Latest)	A React-based framework used for building server-side rendered applications with fast performance and SEO optimization.

	Tools and Technologies	Version	Rationale
Tools and Technologies	Tailwind CSS	Latest	Utility-first CSS framework used for modern UI design and responsive layouts.
	Shadcn/UI	Latest	Component library used to create professional dashboards and UI elements.
	HTML5	Latest	Core markup language used to structure web content.
	CSS3	Latest	Styling language used for layout and responsive design.
	JavaScript	ES6 (Latest)	Scripting language used for frontend logic and interactivity.
	Backend Development		
	FastAPI	Python 3.11+	High-performance backend framework used to build APIs and serve ML models.
	SQLAlchemy	2.0	ORM used for managing database interactions.
	JWT(JSON Web token)	Latest	Used for secure authentication and role-based access control.
	Database		
	PostgreSQL	15+	Relational database used for storing structured IVF and clinical data.
	Machine Learning		
	PyTorch / Keras	Latest	Deep learning frameworks used for embryo grading models.
	XGBoost / TabPFN	Latest	Models used for pregnancy prediction using tabular data.
	Scikit-learn	Latest	Used for preprocessing and anomaly detection.

	Pandas	Latest	Used for data cleaning and analysis.
	NumPy	Latest	Used for numerical computations.
	SHAP	Latest	Used for explainable AI and model interpretation.
	Image Processing		
	OpenCV	Latest	Used for embryo image processing and heatmap generation.
	Testing and Debugging		
	Postman	Latest	Used for testing APIs and debugging backend services.
	UI/UX Design Tools		
	Figma	Free Plan	Used for designing mockups and user interface prototypes.
	DevOps & Deployment		
	Docker / Docker Compose	Latest	Used for containerization and deployment consistency.

1.10 Project Contribution

The Ovulite system introduces several innovative contributions:

- **AI-Driven Embryo Grading:** An advanced machine learning model that accurately grades embryos based on predefined quality parameters, eliminating subjectivity in the process.
- **Predictive Analytics for Pregnancy:** A real-time pregnancy prediction feature that uses AI to forecast the likelihood of pregnancy, assisting veterinarians in making informed decisions.
- **Comprehensive Farm Management Dashboard:** A unified dashboard that helps farm managers monitor and optimize herd reproductive health, improving operational efficiency.

- **Explainable AI (XAI) for Clinical Trust:** Unlike "black-box" systems, Ovulite provides SHAP-based visual explanations for every prediction, showing veterinarians exactly how factors like CL size or blood flow influenced the pregnancy probability.
- **Automated Lab Quality Auditing:** The system introduces an unsupervised anomaly detection loop (Lab QC) that automatically identifies sub-optimal media batches or technician performance drifts without requiring manual oversight.
- **Continuous Learning Feedback Loop:** Ovulite features a self-evolving architecture that automatically triggers model retraining as new pregnancy outcomes are recorded, ensuring the system's accuracy improves over time with specific farm data.

1.11 Relevance to Course Modules

The **Ovulite** project integrates multiple aspects of my academic knowledge and practical skills gained throughout my coursework. Key alignments include:

CSC301 - Design and Analysis of Algorithms Applied algorithmic techniques to optimize the feature selection process and implemented efficient data structures for handling real-time prediction queries within the pregnancy success engine.

CSC270 - Database Systems Designed and implemented a robust **PostgreSQL** relational schema to manage complex IVF-ET data, including donor/sire metadata, recipient physiological records, and a structured registry for embryo image paths.

CSC462 - Artificial Intelligence Developed and trained a **CNN-based (ResNet18)** embryo grading model using image-metadata fusion and utilized **SHAP (Explainable AI)** to provide transparency for veterinarians by highlighting key predictive features like CL size and progesterone levels.

DSC327 - Data Visualization Techniques Used **Pandas** and **Matplotlib** for exploratory data analysis of the DayZee Farms dataset and designed an interactive dashboard in **Next.js** to visualize the IVF funnel, technician performance, and pregnancy success trends.

CSC336 - Web Technologies Built a responsive, glassmorphic web interface using **Next.js** for the frontend and **FastAPI** for a high-performance, Python-native backend to serve real-time machine learning predictions.

DSC415 - Big Data Analytics Applied **Unsupervised Learning (Isolation Forest)** to a large dataset of 488+ clinical records to identify anomalies in lab performance and media batch effectiveness.

CSC291 - Software Engineering Concepts Followed the full SDLC, emphasizing **Modular Design** for the 7 core system modules and **Role-Based Access Control (RBAC)** to ensure secure data management for veterinarians and farm admins.

DSC410 - Topics in Data Science Containerized the entire platform—including the FastAPI backend, Next.js frontend, and PostgreSQL database—using **Docker** to ensure reproducible environments and scalable deployment.

Through Ovulite , I was able to combine theory and practical skills across AI, data science, web development, databases, and cloud deployment, creating a platform that is both academically rigorous and industry relevant.

Chapter 2: Requirement Analysis

This chapter defines the requirements of the Ovulite reproductive intelligence system by translating high-level project objectives into clear system specifications. It identifies user classes, describes system functionality, and establishes measurable non-functional requirements such as performance, reliability, and security. These requirements define the system's expected behavior and provide a foundation for design, implementation, and validation.

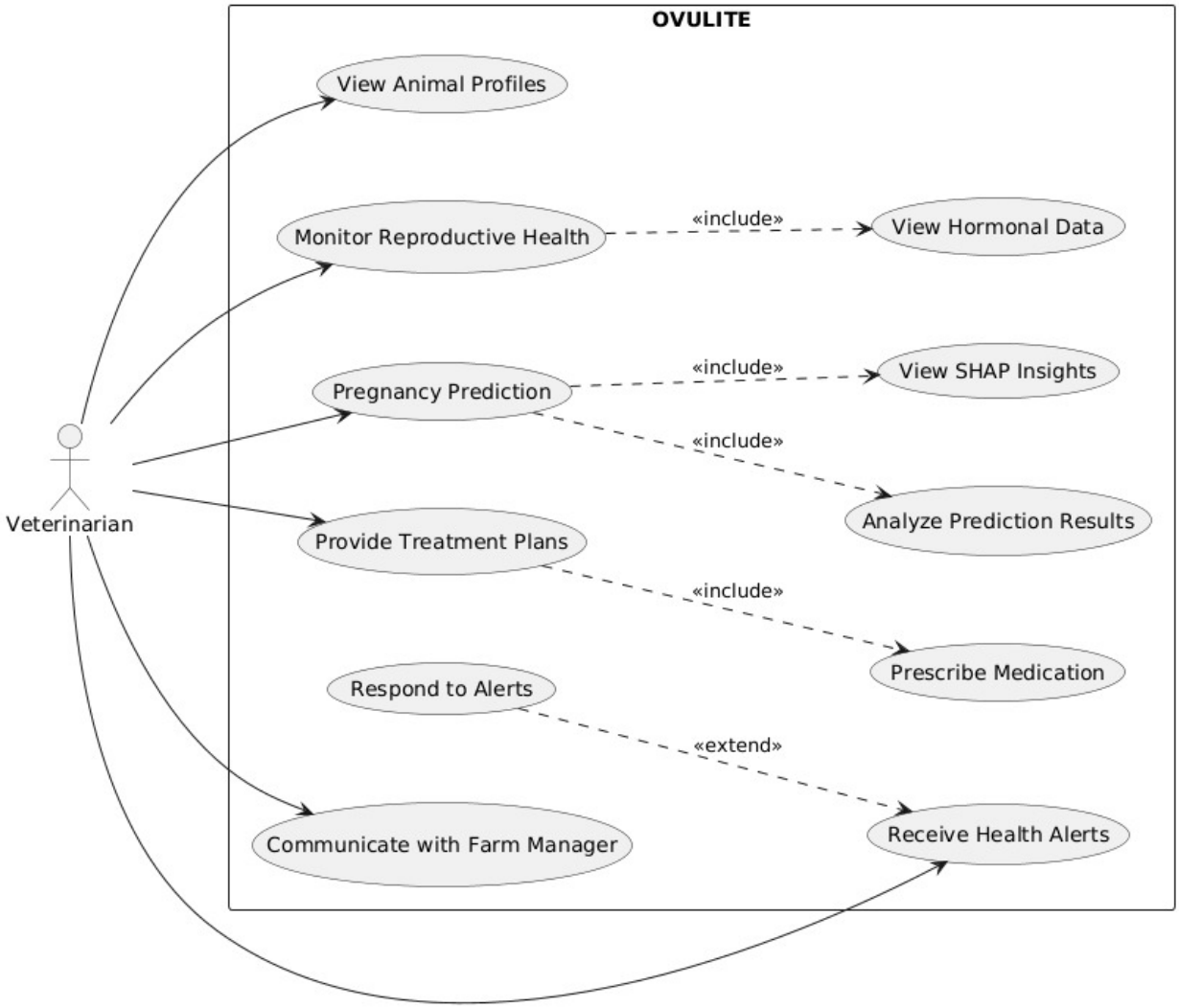
2.1 User Classes and Characteristics for Ovulite Project

Table 3 Ovulite userclass and Characteristics

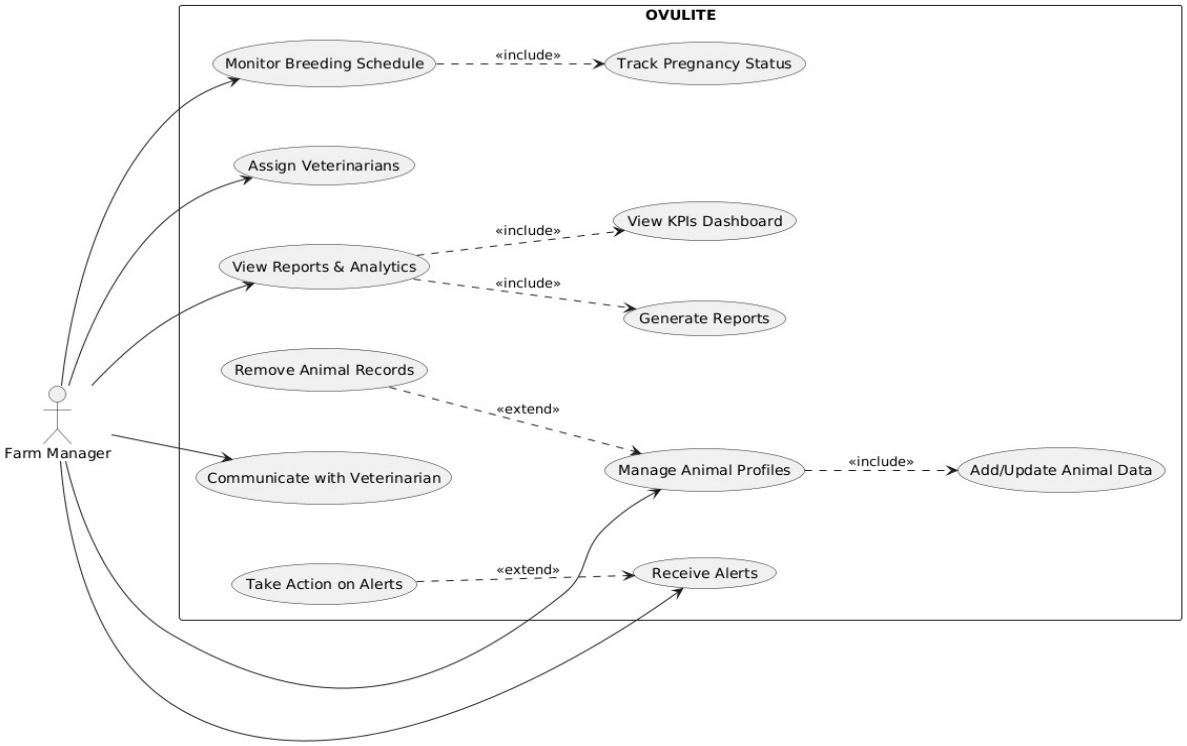
User Class	Description
Veterinarian	Uses the system to input recipient data, monitor reproductive parameters, request pregnancy predictions, and record clinical outcomes.
System Administrator	Manages user accounts, system configuration, access control, and ensures system security and data integrity.
Farm Manager	Monitors reproductive performance using dashboards and analytics, and uses system insights for decision-making.
Embryologist / IVF Lab Technician	Handles embryo-related operations including image uploads, grading, and maintaining laboratory data within the system.

2.1.1 Use case Diagram

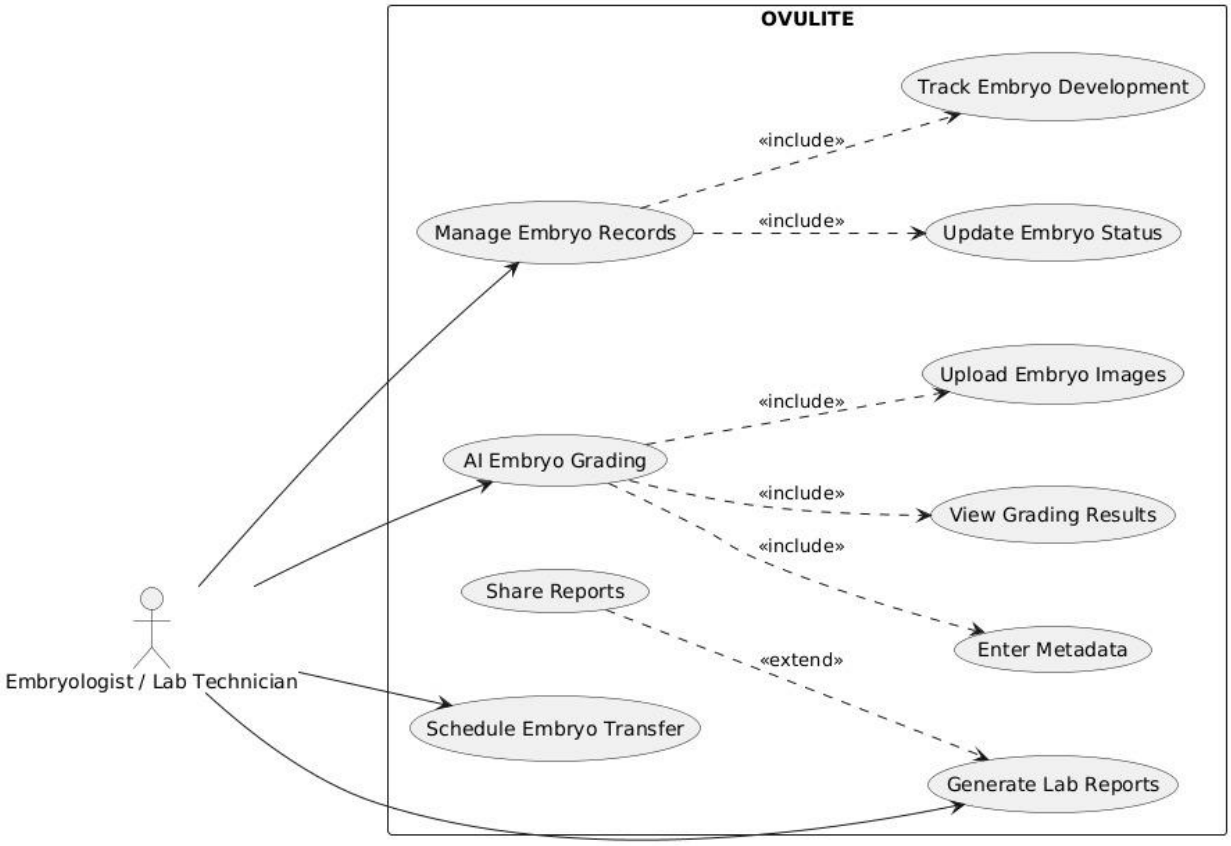
2.1.1.1 Actor: Veterinarian



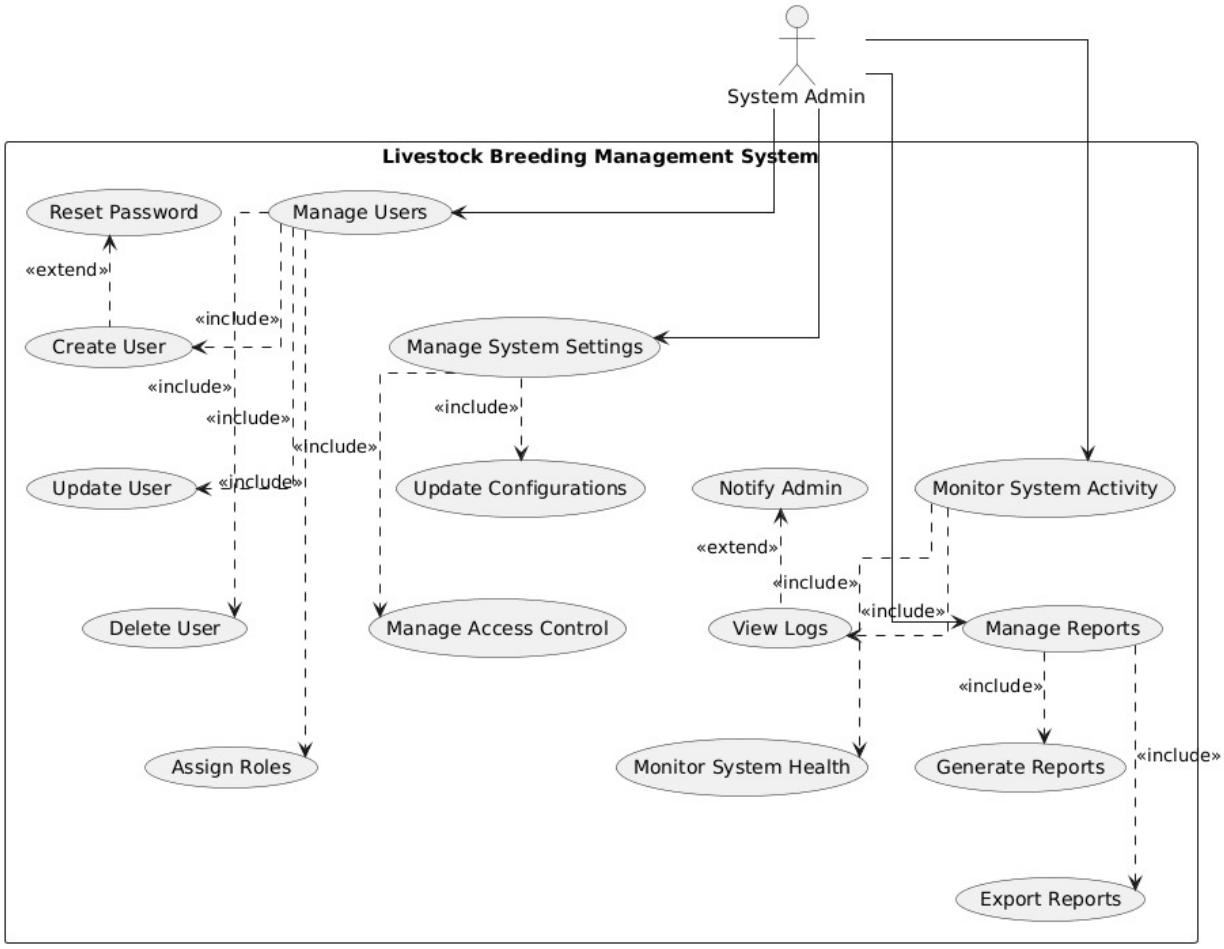
2.1.1.2 Actor: Farm Manager



2.1.1.3 Actor: Embryologist



2.1.1.4 Actor: System Adminitrator



2.1.2 Requirement Identifying Technique

The functional requirements of the Ovulite system are identified using **Mockup-Based Requirement Analysis**. This approach derives system requirements directly from user interface designs, ensuring that all requirements are user-centered, traceable, and aligned with system functionality.

In this method:

1. **Mock-up Creation:** Key system features are divided into user interface screens such as dashboards, prediction modules, and data entry forms. Mockups are created to represent these interactions visually.
2. **Visual Workflow Design:** The sequence of user interactions across different screens is defined to ensure logical navigation and usability.
3. **Element Analysis:** Each interactive element within the mock-ups (buttons, input fields, charts, and navigation links) is thoroughly analyzed. This detailed analysis ensures that every element within the UI is aligned with the platform's functionality.

4. **Requirement Derivation:** For each mock-up and corresponding interactive element, the expected system behavior is identified and documented. These behaviors are written as Functional Requirements (FRs) with unique identifiers, such as FR1.1 or FR3.2, to ensure clear traceability and the ability to test functionality later.
5. **Structured Documentation:** All derived functional requirements are organized into clear, structured tables, each corresponding to its originating mock-up. This structured approach ensures that every requirement is easy to track and implement, reducing ambiguity during the development process.
6. **Business Rule Definition:** Constraints and validation rules are defined to enforce correct system behavior and maintain data consistency.

2.1.3 Backend Process Requirements

The Ovulite backend utilizes microservices architecture, specifically designed to support the platform's core promises of scalability, automation, and real-time operations. The backend orchestrates multiple processes, including data collection, AI model execution, and campaign management workflows, allowing for intelligent, data-driven insights while minimizing manual intervention.

These processes are outlined using an Event-Response model, which clearly defines how the system should respond to specific events. Below are the detailed backend process requirements:

Table 4 Event Response Table

Event	System State	Response
User logs in	Authentication Module	Validates user and grants role-based access
User enters IVF/ET data	Data Intake & Validation	Validates inputs and stores structured data
ET data submitted	Prediction Engine	Triggers pregnancy prediction model
Prediction requested	Prediction Engine	Displays probability with contributing factors
Embryo image uploaded	AI Embryo Grading	Generates embryo grade with confidence
Lab data updated	LabQC Module	Analyzes performance across batches, media, technicians
Anomaly detected	LabQC Module	Generates QC alert for abnormal patterns
User accesses dashboard	Dashboard Interface	Displays KPIs, trends, and alerts
Protocol analysis requested	Protocol Analytics Module	Provides protocol effectiveness insights
Pregnancy outcome recorded	Outcome Tracking Module	Updates case record with outcome data
New outcome data available	Learning Module	Updates model performance and learning

These **backend process requirements** ensure that the **Ovulite** system remains scalable, secure, and highly available while providing real-time decision-making capabilities for users. Each event is clearly mapped to its corresponding response, allowing the system to operate efficiently and autonomously.

2.2 Functional Requirements

This section outlines the functional requirements derived from user interactions and mockups for the Ovulite system. Each requirement is documented at the feature level (e.g., User Login, Embryo Grading) and includes a unique identifier (e.g., FR1.1, FR1.2), a clear, testable statement, reference to the corresponding mockup or UI element, and optional Business Rules that define domain-specific behaviors. These business rules ensure alignment with organizational policies or legal constraints while clarifying system behavior.

2.2.1 Mockups:

2.2.1.1 Mockups M1 – Login Page Screen

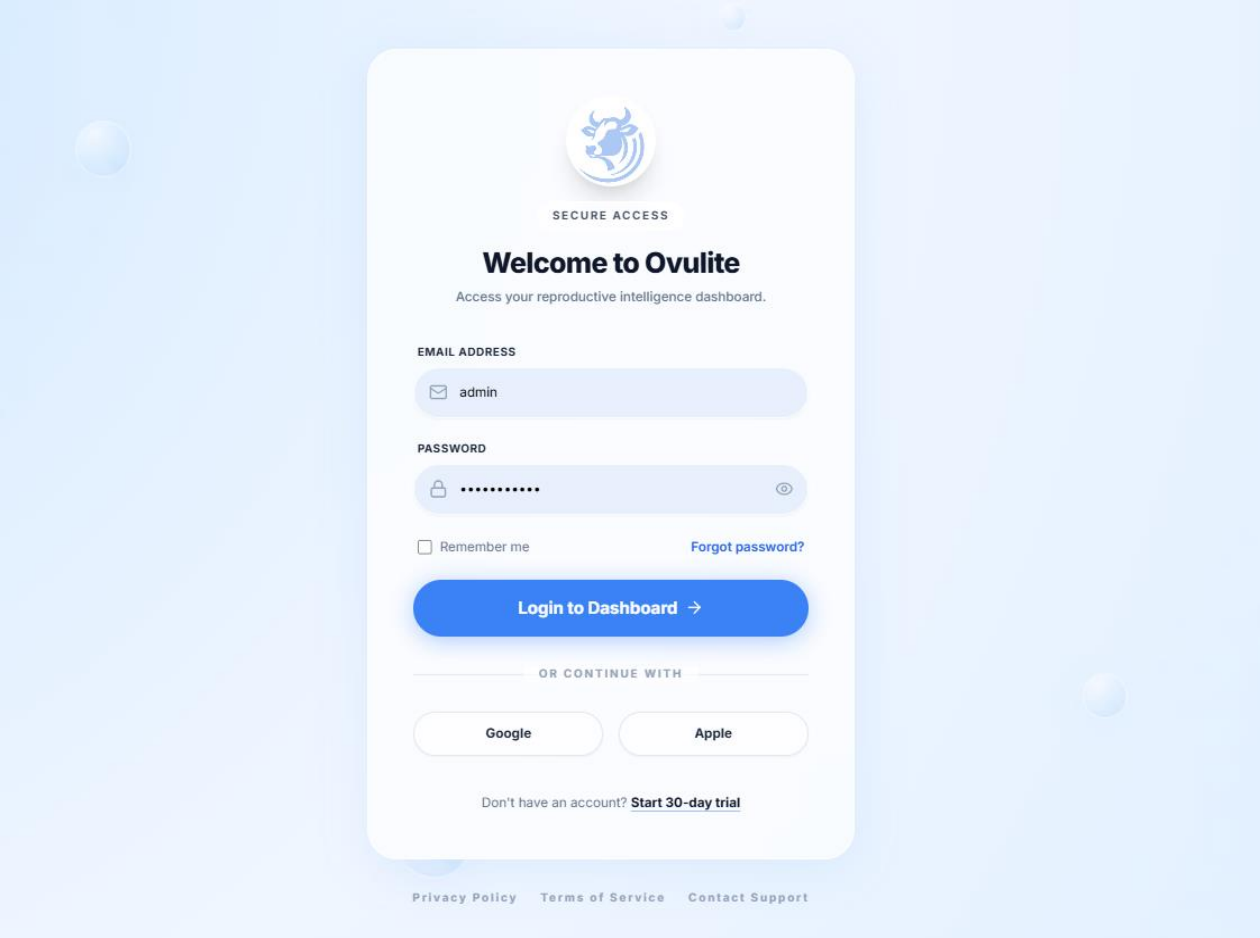


Figure 1 Mockup M1 for Login Page

Functional Requirements Derived from Mockup M1 – User Authentication (Login)

Table 5 Functional Requirements Derive from Mockup M1

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
User Login	FR-1.1: The system shall allow users to securely log in with email and password credentials.	The email format should follow the standard pattern (user@example.com), and the password should contain at least 8 characters.
Forgot Password	FR-1.2: The system shall allow users to reset their password via a password recovery process.	Only registered users can initiate password recovery, and the system must

		send a password reset link to the registered email.
User Authentication	FR-1.3: The system shall enforce strong authentication rules, including blocking users after 5 failed login attempts.	Account access should be locked after 5 failed attempts to prevent brute-force attacks and require manual unlock by admin.
Remember Me Functionality	FR-1.4: The system shall offer a 'Remember Me' option to keep the user logged in across sessions.	The Remember Me option should store the session for a configurable period (e.g., 7 days) for easier user access.
Session Timeout	FR-1.6: The system shall automatically log users out after 15 minutes of inactivity for security.	The system must clear the session and redirect the user to the login page after the specified inactivity timeout period.

2.2.1.2 Mockup M2 –Dashboard Page

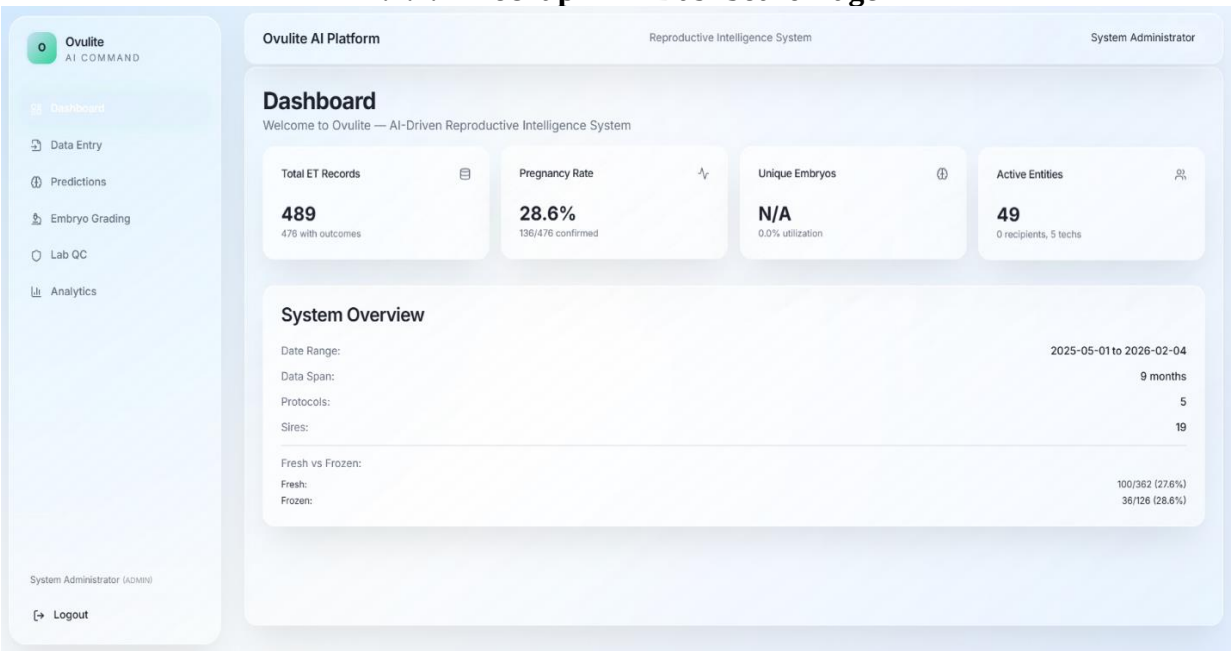


Figure 2 Mockup M2 for Dashboard Page

Functional Requirements Derived from Mockup M2- Dashboard

Table 6 Functional Requirements Derive from Mockup M2

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Total ET Records Display	FR-2.1: The system shall display total embryo transfer records (ET) and outcomes in the dashboard.	ET records must be updated in real-time, and the system must provide a filter for viewing records by outcome or protocol.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Pregnancy Rate Display	FR-2.2: The system shall show the pregnancy rate calculated based on the total transfer records.	The pregnancy rate must be computed from successful outcomes divided by the total transfers, displayed as a percentage.
Unique Embryos	FR-2.3: The system shall display the number of unique embryos, along with their utilization rate.	Embryo utilization should be calculated based on the number of embryos successfully used in transfers divided by the total available embryos.
Active Entities Count	FR-2.4: The system shall display the total number of active entities, including recipients and technicians.	Active entities should reflect the current number of participants, including recipients and technicians, updated daily.
System Overview (Date Range & Data Span)	FR-2.5: The system shall allow users to view a summary of the date range and data span for ongoing analytics.	Date range and data span should be displayed clearly and should be adjustable by the user for comparative analysis of different periods.
Fresh vs Frozen Data	FR-2.6: The system shall present the comparison between fresh and frozen embryos in terms of transfer success.	Success rates for both fresh and frozen transfers should be calculated and displayed with a percentage comparison for easy analysis.

2.2.1.3 Mockup M3– ET Transfer Records

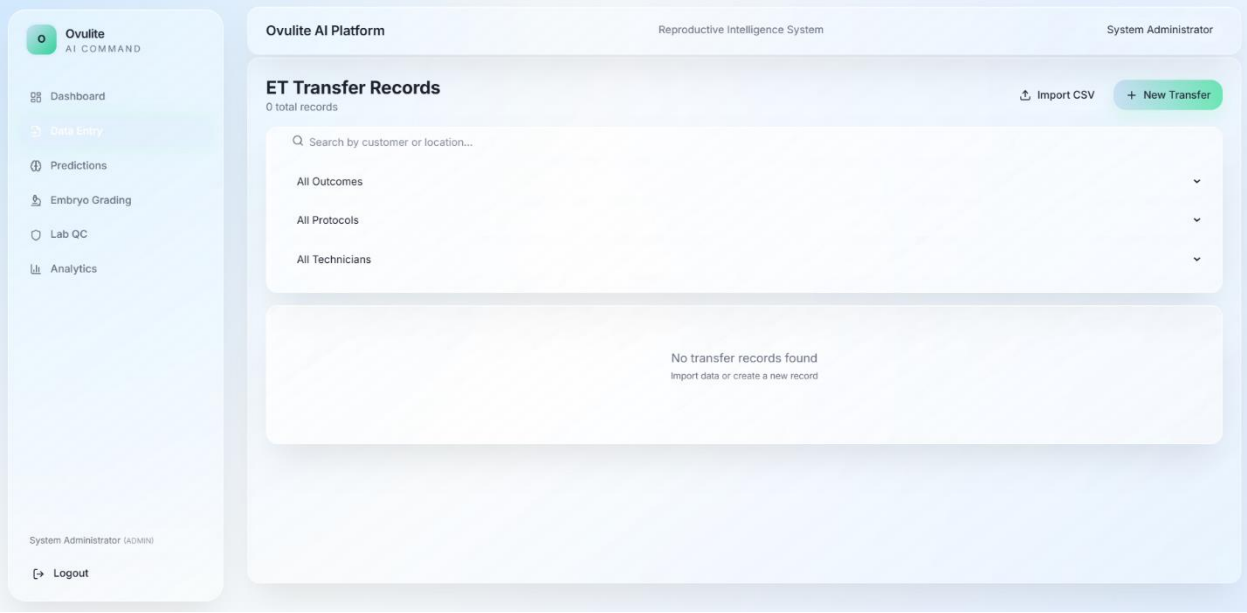


Figure 3 Mockup M3 for ET Transfer Records

Functional Requirements Derived from Mockup M3- ET Transfer Records

Table 7 Functional Requirements Derived from Mockup M3

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
ET Transfer Records Display	FR-3.1: The system shall display all ET transfer records in the data entry section.	ET records should be updated in real-time and must display details such as outcome, protocol, and technician for each record.
Filter ET Transfer Records	FR-3.2: The system shall allow users to filter ET transfer records by outcome, protocol, or technician.	Users must be able to filter records by multiple parameters (outcome, protocol, technician), and the system should update the results within 5 seconds.
CSV Import	FR-3.3: The system shall allow users to import ET transfer records using CSV files.	The CSV import feature must support batch uploads, check for duplicates, and validate the data format before importing.
New Transfer Record Entry	FR-3.4: The system shall provide an option to add new ET transfer records through a manual form.	New record entry must include required fields such as ET number, customer, technician, and transfer date.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
No Transfer Records Found Notification	FR-3.5: The system shall display a message if no ET transfer records exist, prompting the user to add new records.	When no records exist, the message should read "No transfer records found" with a call to action for creating a new record or importing data.
Export Transfer Records	FR-3.6: The system shall provide an export option to download ET transfer records in CSV format.	Users should be able to export filtered or all records as a CSV file, with no data loss during export.

2.2.1.4 Mockups M4 – Reports & Export

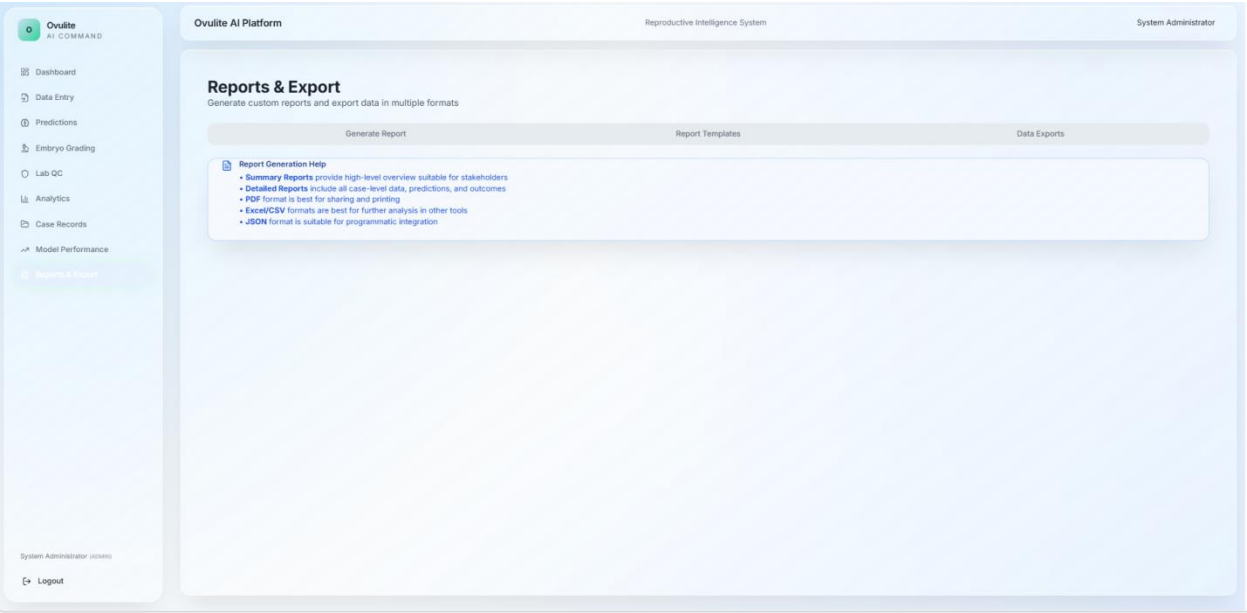


Figure 4 Mockup M4 for Reports & Export

Functional Requirements Derived from Mockup M4- Reports & Export

Table 8 Functional Requirements Derived from Mockup M4

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Report Generation	FR-4.1: The system shall allow users to generate reports based on selected data.	Reports must include summaries and detailed analysis.
Report Templates	FR-4.2: The system shall provide predefined report templates.	Templates must be standardized.
Export Formats	FR-4.3: The system shall allow export in PDF, Excel, and JSON formats.	Export must preserve data integrity.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Export Data	FR-4.4: The system shall allow users to export analytics and case data.	Export must include timestamps and metadata.
Report Help Section	FR-4.5: The system shall display guidance for report generation.	Help must explain format usage.

2.2.1.5 Mockups M5 – Case Records & Traceability

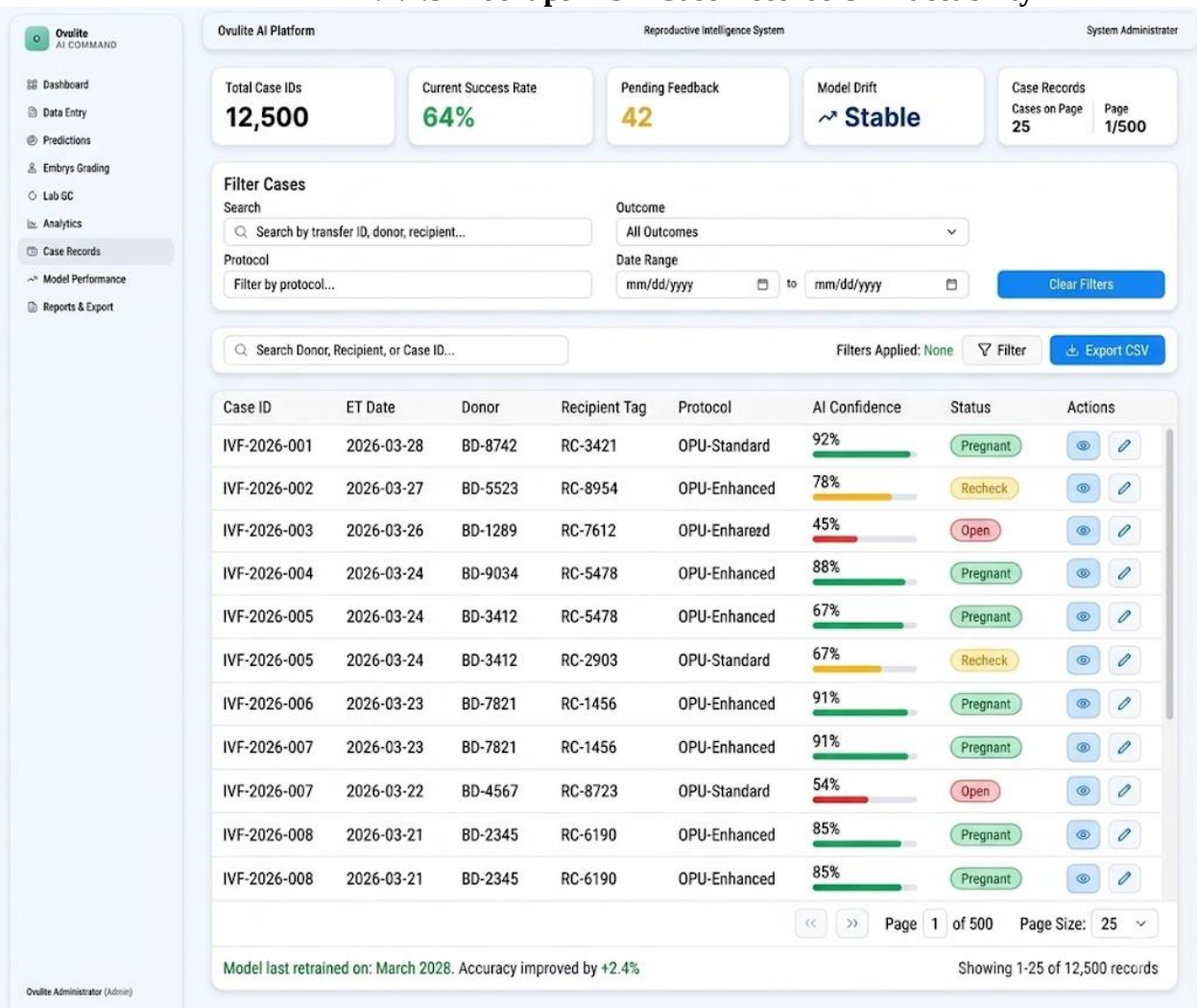


Figure 5 Mockup M5 for Case Records & Traceability

Functional Requirements Derived from Mockup M5- Case Records & Traceability

Table 9 Functional Requirements Derived from Mockup M5

Feature(derived from UI)	Functional Requirement(FR-ID: Statement)	Business Rule
Case Search	FR-5.1: The system shall allow users to search cases using donor ID, recipient ID, or transfer ID.	Search must support partial matching.
Case Filtering	FR-5.2: The system shall allow filtering based on protocol, outcome, and date range.	Filters must be combinable.
Case Pagination	FR-5.3: The system shall display cases with pagination controls	Page size must be adjustable.
Case Summary Metrics	FR-5.4: The system shall display summary metrics such as total cases and current page.	Metrics must update dynamically
Case Detail Access	FR-5.5: The system shall allow users to open individual case records.	Access must be role-based.
Error Handling Display	FR-5..6: The system shall display error messages if data retrieval fails	Errors must be clearly visible.

2.2.1.6 Mockup M6- Embryo Grading

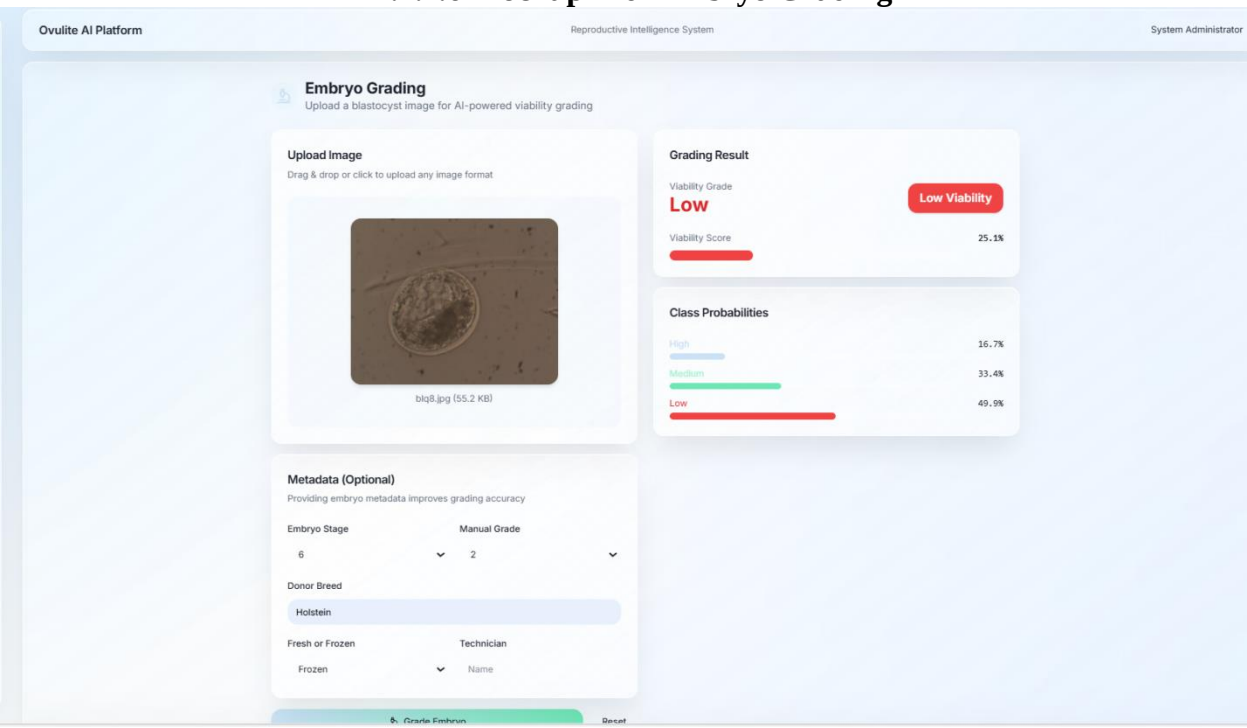


Figure 6 Mockup M6 for Embryo Grading

Functional Requirements Derived from Mockup6 - Embryo Grading

Table 10 Functional Requirements Derive from Mockup M6

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Embryo Image Upload	FR-6.1: The system shall allow users to upload embryo images in any format (JPEG, PNG, etc.), with a maximum size of 10MB.	Embryo images should be uploaded by dragging and dropping or clicking the upload area. File format validation should be performed before the upload.
Metadata Input	FR-6.2: The system shall allow users to enter metadata for embryo grading, such as Embryo Stage, Manual Grade, Donor Breed, etc.	Metadata fields (e.g., Embryo Stage, Manual Grade) should be optional but will improve grading accuracy.
Embryo Grading Result Display	FR-6.3: The system shall display the grading result for the embryo and AI-powered viability score.	Grading results must be calculated and clearly displayed, with the AI score showing the predicted viability of the embryo.
Embryo Image & Grading Feedback	FR-6.4: The system shall provide AI feedback along with the grading results for the uploaded embryo image.	The feedback should include visual indicators like heatmaps that highlight important features contributing to the grading score.
No Result Display	FR-6.5: If no grading result is available, the system shall display a message, "No result yet", prompting users to upload an embryo image for grading.	The AI result should be displayed within 5 seconds of uploading the image, or else an appropriate message should be shown.
Upload Confirmation	FR-6.6: The system shall show a confirmation message when the image is successfully uploaded, and the AI model is ready for grading.	Once the embryo image is uploaded, a successful confirmation message should indicate that the image is ready for processing and grading.

2.2.1.7 Mockup M7- Lab QC & Anomaly Detection

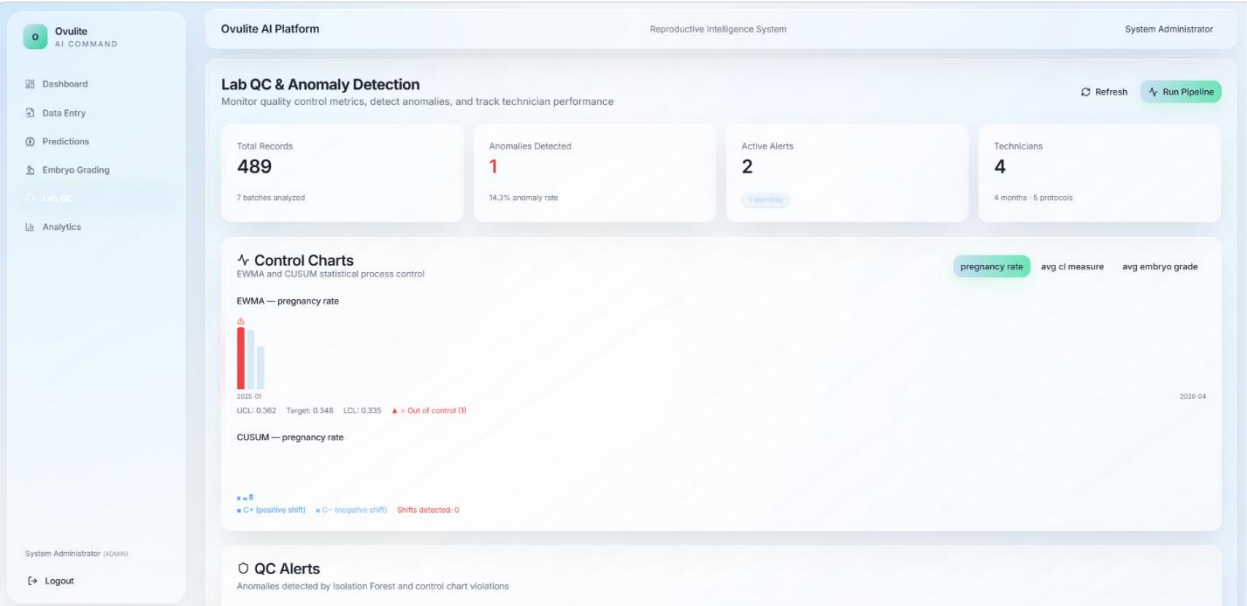


Figure 7 Mockup M7- - Lab QC & Anomaly Detection

Functional Requirements Derived from Mockup7 - Lab QC & Anomaly Detection

Table 11 Functional Requirements Derived from Mockup M7

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Lab QC & Anomaly Detection Overview	FR-7.1: The system shall display total records and anomaly detection metrics (e.g., 14.3% anomaly rate).	Anomalies should be detected using advanced algorithms (e.g., Isolation Forest or One-Class SVM) and displayed prominently.
Control Chart Visualization	FR-7.2: The system shall provide control charts (e.g., EWMA, CUSUM) to track performance over time.	Control charts must display statistical process control information with real-time updates and highlight any out-of-control points.
Alert Notifications	FR-7.3: The system shall display active alerts (e.g., warnings) that are triggered by QC issues or anomalies.	Active alerts should be prominently displayed on the dashboard with the ability to click through for more details.
Technician Performance Tracking	FR-7.4: The system shall track technician performance and display key metrics, including their associated protocols.	Technician performance should be linked to their outcomes, and any significant changes in performance should trigger alerts.
Real-Time Data Updates	FR-7.5: The system shall update all performance metrics, charts, and alerts in real-time, reflecting the most recent data.	The system must update the pregnancy rate, CL measure, and embryo grade in near real-time, typically within 30 seconds.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Anomaly Detection Metrics	FR-7.6: The system shall track and flag anomalies in lab performance (e.g., pregnancy rate deviations).	Anomalies should be analyzed and reported for each batch, and deviations must be flagged for further investigation by lab operators.
Pregnancy Rate Tracking	FR-7.7: The system shall allow users to monitor the pregnancy rate as a key metric in the control chart section.	The pregnancy rate should be a dynamic metric that updates regularly and reflects the current batch's success rate.

2.2.1.8 Mockup M8 - Data Entry - New ET Transfer

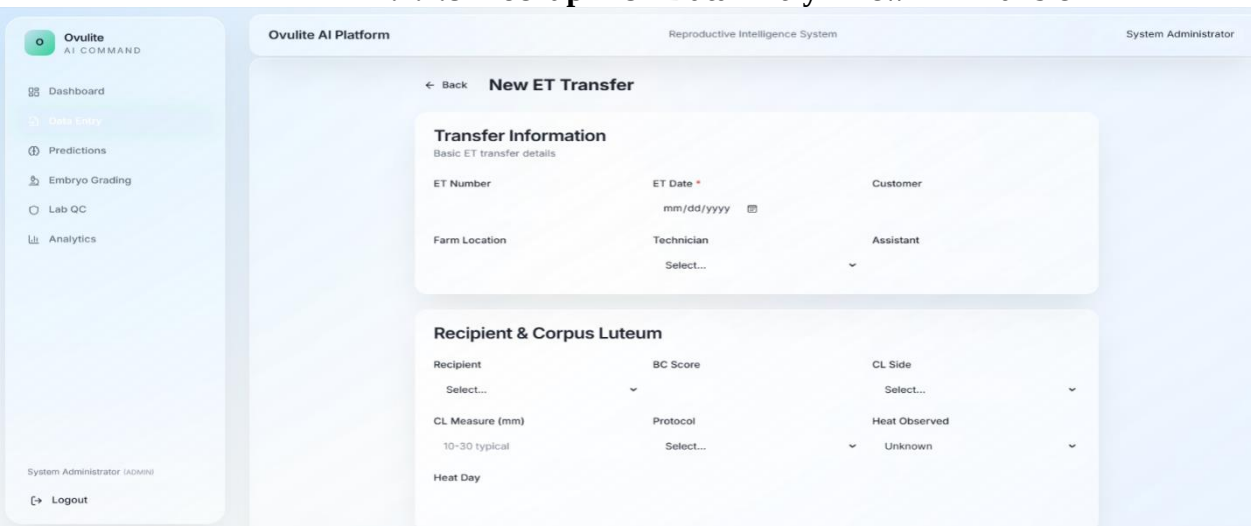


Figure 8 Mockup-8 Data Entry - New ET Transfer

Functional Requirements Derived from Mockup8 - New ET Transfer

Table 12 Functional Requirements Derived from Mockup 8

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
ET Transfer Information Entry	FR-8.1: The system shall allow users to enter basic ET transfer details such as ET number, ET date, and customer information.	ET number and ET date are mandatory fields, while customer and technician information should be linked to existing records.
Technician Selection	FR-8.2: The system shall provide a drop-down menu for selecting the technician responsible for the ET transfer.	The technician list should be populated from a pre-defined list of available technicians, with the option to select or update as needed.
CL (Corpus Luteum) Measure	FR-8.3: The system shall allow users to input CL measure (in mm) from 10-30 mm typical values.	CL measure should accept numerical input only, and a validation rule should ensure it falls within a valid range (10-30 mm).
Recipient Information	FR-8.4: The system shall provide a drop-down menu to select the recipient of the ET transfer.	Recipient data must be pulled from existing records for consistency, and should be selectable from the available list of recipients.
BC Score Entry	FR-8.5: The system shall allow users to input BC score for the recipient (optional).	BC score should accept numerical values, with a range check to ensure valid input.
Protocol Selection	FR-8.6: The system shall provide a drop-down menu for selecting the protocol used for the ET transfer.	Protocol should be selected from a predefined list, ensuring consistency in selecting valid protocols for each transfer.
Heat Observed Selection	FR-8.7: The system shall allow users to select whether heat was observed during the transfer process.	The Heat Observed field should offer options like Unknown, Yes, or No, with a default value of Unknown if no selection is made.

2.2.1.9 Mockup M9- Pregnancy Prediction

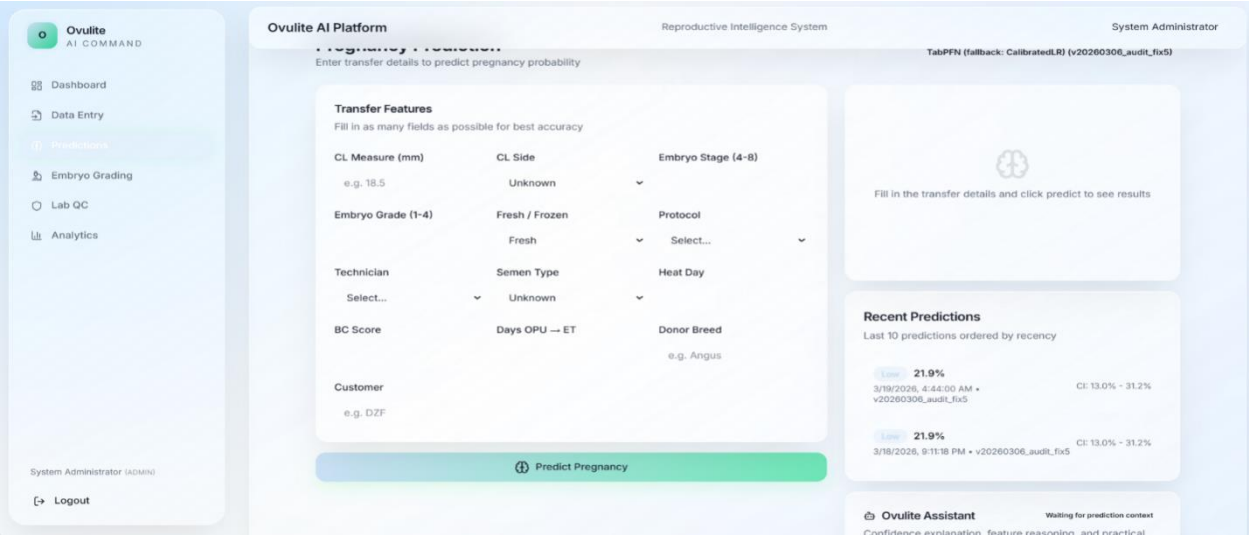


Figure 9 Mockup-9 Pregnancy Prediction

Functional Requirements Derived from Mockup9 - Pregnancy Prediction

Table 13 Functional Requirements Derived from Mockup-9

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Transfer Features Input	FR-9.1: The system shall allow users to input key transfer features such as CL measure, embryo grade, and protocol to predict pregnancy.	Users must provide as many details as possible for higher prediction accuracy (e.g., embryo grade, CL measure, semen type).
Predict Pregnancy Button	FR-9.2: The system shall allow users to click a button ("Predict Pregnancy") to generate a pregnancy prediction based on the input features.	The Predict Pregnancy button should trigger the system to use AI models to calculate pregnancy probabilities and display results.
Recent Predictions Display	FR-9.3: The system shall display recent predictions in a scrollable list, showing the last 10 predictions ordered by recency.	The system must display the date, pregnancy prediction, and confidence intervals (CI) for the last 10 predictions to maintain transparency.
Data Fields for Prediction	FR-9.4: The system shall allow users to input additional relevant data for prediction, such as technician, BC score, semen type, and donor breed.	Each input field must allow data entry in appropriate formats (e.g., BC score as numeric, semen type from a dropdown).
Confidence Interval Display	FR-9.5: The system shall display the confidence interval (CI) for each pregnancy prediction to provide an estimation of prediction reliability.	Confidence interval (CI) should be shown as a range, for example, 13.0% - 31.2%, alongside the pregnancy rate prediction.

Prediction Accuracy Enhancement	FR-9.6: The system shall highlight the most recent prediction and allow users to view detailed results for improved insights into model performance.	The recent predictions should be clearly listed with actionable insights, like confidence or pregnancy success rate.

2.2.1.10 Mockup M10-Analytical Dashboard

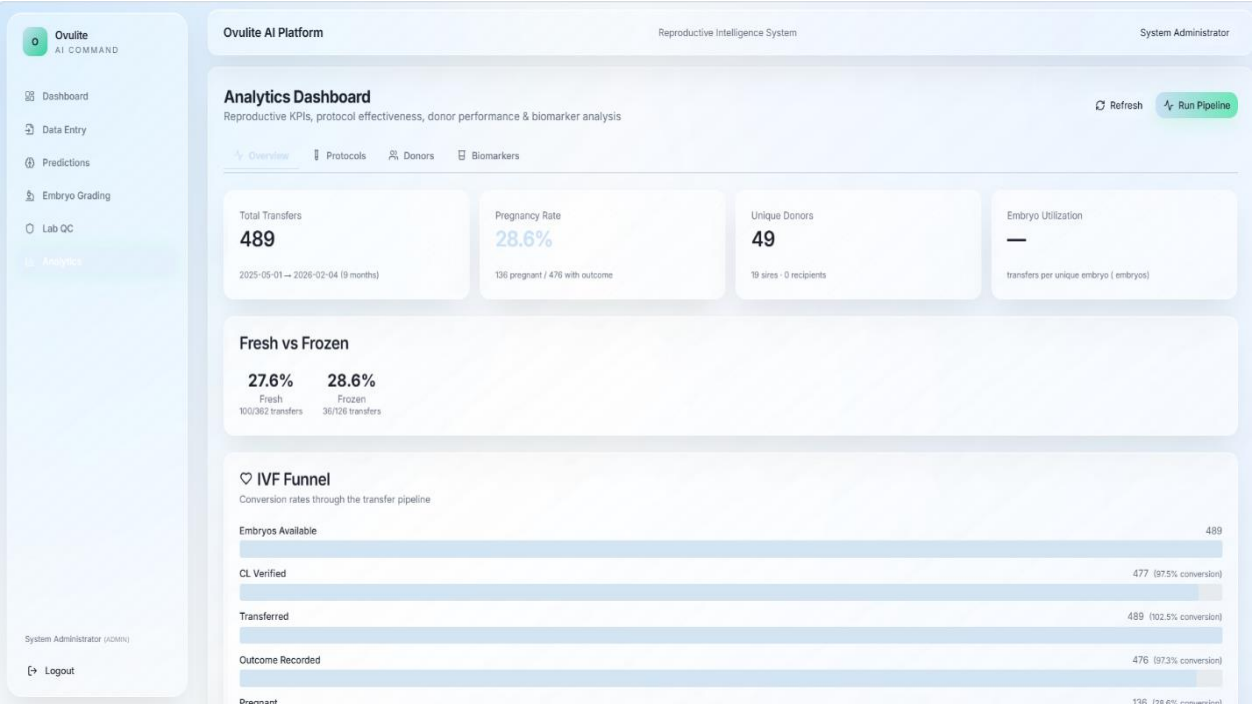


Figure 10 Mockup-10 Analytical Dashboard

Functional Requirements Derived from Mockup10 - Analytical Dashboard

Table 14 Functional Requirements Derived from Mockup-10

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Total Transfer Records Display	FR-10.1: The system shall display the total number of embryo transfers in the dashboard, including success rates.	Total transfer records should be updated in real-time and show total transfers and the number of transfers with outcomes.
Pregnancy Rate Display	FR-10.2: The system shall calculate and display the pregnancy rate based on the total number of transfers and pregnancy outcomes.	Pregnancy rate must be displayed as a percentage and must automatically update with each new data entry, based on successful outcomes.
Fresh vs Frozen Comparison	FR-10.3: The system shall display the comparison between fresh and frozen transfers in terms of success rates.	The system must calculate the fresh vs frozen transfer success rate, allowing users to evaluate the performance of both types of transfers.

Feature (derived from UI)	Functional Requirement (FR-ID: Statement)	Business Rule
Unique Donors Display	FR-10.4: The system shall display the number of unique donors utilized during the time period shown in the system overview.	Unique donors should be calculated and displayed by tracking the sire/recipient combinations, and donor data must be refreshed regularly.
IVF Funnel Metrics	FR-10.5: The system shall display conversion rates at each stage of the IVF process, such as embryos available, transferred, and outcome recorded.	The conversion rates should be visualized in a clear, easy-to-understand manner, tracking progress from embryo availability to pregnancy outcome.
Data Time Span and Date Range Display	FR-10.6: The system shall allow users to view the date range for the data and track the data span of IVF activities.	The system must show start and end dates (for example, 2025-05-01 to 2026-02-04) and allow users to filter data by date ranges.
Embryo Utilization	FR-10.7: The system shall track embryo utilization, showing the number of transfers per unique embryo.	Embryo utilization rate must reflect the ratio of embryos transferred to embryos available, ensuring proper tracking and use of embryos.

2.2.1.11 Mockups M11 – Analytics Dashboard(Protocol+ Donors + Biomarkers)

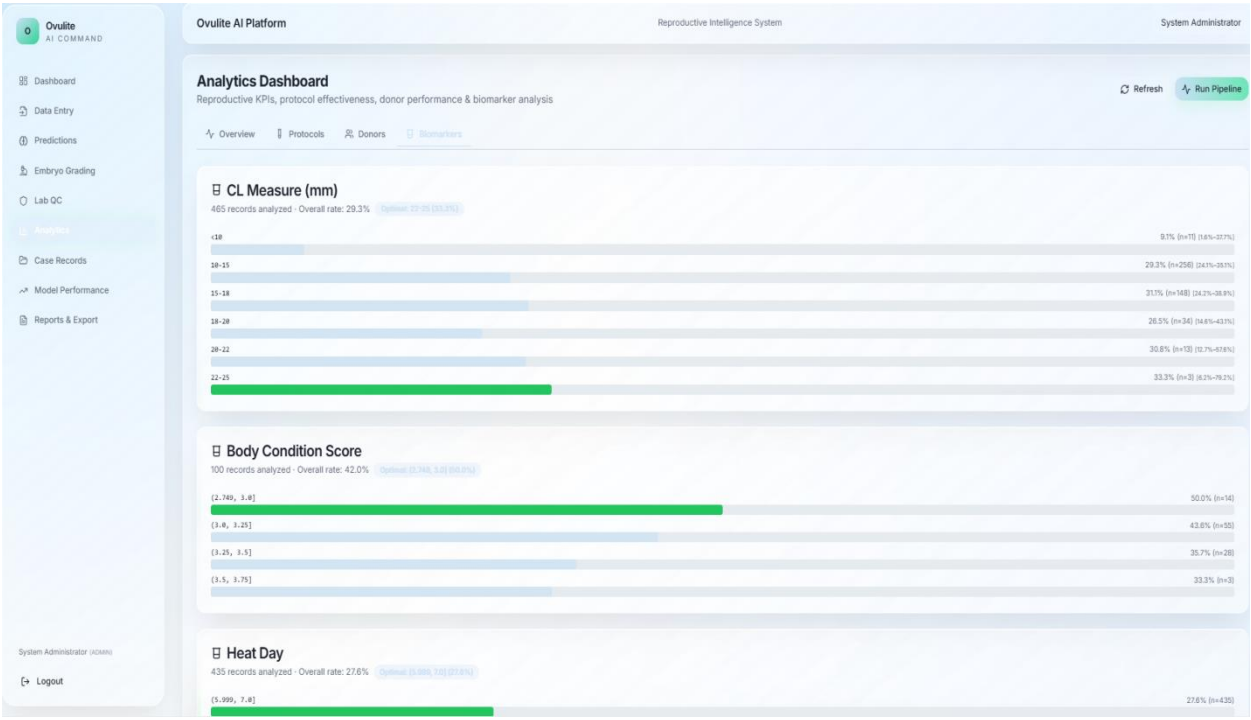


Figure 11 Mockup 11- Analytics Dashboard

Functional Requirements Derived from Mockup 11 - Analytics Dashboard (Protocol+ Donors + Biomarkers)

Table 15 Functional Requirements Derived from Mockup 11

Feature(Derived from UI)	Functional Requirement	Business Rules
Protocol Pregnancy Rates	FR-11.1: The system shall display pregnancy rates for each synchronization protocol.	Rates must include total transfers and successful pregnancies.
Confidence Interval Display	FR-11.2: The system shall display 95% confidence intervals for each protocol.	CI must be shown alongside each rate.
Protocol Comparison Bars	FR-11.3: The system shall visualize protocol performance using comparative bar charts.	Color coding (green/yellow/red) must indicate performance levels
Protocol Details Table	FR-11.4: The system shall display detailed protocol data including transfers, pregnancies, and rates.	Table must be sortable and structured.
Donor Performance Ranking	FR-11.5: The system shall display donor performance ranked by pregnancy success rate.	Ranking must be based on historical outcomes.
Donor Details Table	FR-11.6: The system shall display donor-level data including breed, transfers, and pregnancy rate.	Each donor must have traceable records.

2.3 Non-Functional Requirements

This section specifies the non-functional requirements (NFRs) that define the quality attributes of the Ovulite platform, an AI-powered solution for bovine embryo transfer management.

2.3.1 Reliability

REL-1: The Ovulite platform shall target 99.5% availability during operating hours (8 AM – 8 PM local time) to ensure continuous access to embryo grading and pregnancy prediction functionalities.
REL-2: The system shall handle external API failures (e.g, external data sources external data sources or analytics services) by notifying users of limited functionality and retrying data collection once the service is restored. Target recovery window: <5 seconds

REL-3: In case of system failure or restart, critical data (embryo grades, health metrics) shall be preserved and the platform shall resume from the most recent stable state.
REL-4: The system shall implement offline caching mechanisms to retain critical data (e.g., embryo grading**,** pregnancy prediction) during short connectivity losses, ensuring data is available upon reconnection. Cached data should load within 10 seconds**.**

2.3.2 Usability

USE-1: A new user shall be able to complete account registration and view the demo dashboard within **60 seconds**.
USE-2: From the main dashboard, users shall access key insights (e.g., embryo grading results, pregnancy predictions) in 3 clicks or fewer.
USE-3: The **Ovulite Assistant** (Chatbot) shall provide a response to clinical queries within **3 seconds**.
USE-4: Insights (e.g., pregnancy prediction) shall be displayed in plain language (e.g., "This embryo has a 90% chance of pregnancy"), free of technical jargon.
USE-5: After receiving an alert from the Protocol Effectiveness Analysis, users shall be able to filter protocol effectiveness reports in ≤ 3 interactions.

2.3.3 Performance

PER-1: Embryo grading and pregnancy prediction data visualizations shall update in near real-time, with a latency of ≤ 5 seconds after new data is processed.
PER-2: The Embryo Grading System shall process an embryo image (inference) within **3-5 seconds**
PER-3: The Pregnancy Prediction Engine shall provide results within **2 seconds** after receiving tabular input (CL size, P4)

2.3.4 Security

SEC-1: User credentials shall be stored securely using bycrypt hashing algorithms.
SEC-2: External integrations (e.g., APIs, third-party services) shall use OAuth-based token authorization to securely connect user accounts.
SEC-3: Sessions shall automatically expire after 30 minutes of inactivity to prevent unauthorized access.
SEC-4: The system shall comply with applicable data protection regulations relevant to deployment region.
SEC-5: All data transmission between the client and server shall be encrypted using HTTPS (TLS 1.2+) to protect against data breaches.

2.3.5 Accessibility

ACC-1: The system shall comply with WCAG 2.1 accessibility guidelines, ensuring that the platform is usable by individuals with disabilities.

ACC-2: The UI shall support keyboard navigation and be compatible with screen readers (e.g., JAWS, NVDA).
ACC-3: Contrast ratios between text and background should be compliant with accessibility standards (at least 4.5:1).

2.3.6 Portability

PORT-1: The system shall be containerized using **Docker Compose**, orchestrating the FastAPI backend, Next.js frontend, and PostgreSQL database into a single deployable unit.
PORT-2: The platform should support cross-platform compatibility, allowing deployment on both Windows and Linux environments.
PORT-3: The system shall be compatible with mobile operating systems (e.g., iOS, Android) for mobile web or app deployment.

2.3.7 Maintainability

MAINT-1: The platform shall have a modular architecture, so updates or fixes in one module (e.g., AI Embryo Grading) do not require changes to other modules.
MAINT-2: The system shall follow a modular architecture enabling independent updates to system components.
MAINT-3: Security patches or critical updates from third-party tools or frameworks shall be integrated, tested, and deployed within 48 hours of release.
MAINT-4: Documentation of the codebase shall be maintained with clear instructions for adding new features or fixing bugs.

2.4 External Interface Requirements

The Ovulite platform will be accessed through a web-based graphical user interface (GUI), designed to be intuitive, user-friendly, and efficient, catering to veterinarians, farm managers, and lab technicians with varying levels of technical expertise.

2.4.1 User Interfaces Requirements

UI-1: The user interface shall maintain a consistent and intuitive design across all modules, including embryo grading, pregnancy prediction, real-time health monitoring, and user management. This includes the main dashboard, task-specific screens, and all configuration pages.
UI-2: The Embryo Grading Dashboard shall present embryo quality data and viability scores in a clear, visually digestible format, enabling veterinarians to make quick, informed decisions regarding embryo transfer.
UI-3: The platform's interface shall be responsive, ensuring full functionality and a clean layout across a variety of screen resolutions (desktop, laptop, tablet, and mobile devices).

UI-4: Key navigational elements (e.g., dashboard link, health data alerts, Embryo Grading quick access) shall be persistently visible and easily accessible from all primary views within the application.
UI-5: The system shall support interactive elements within its dashboards. For example, users shall be able to hover over embryo quality heatmaps and pregnancy prediction graphs to view specific data points and trends for quick analysis.
UI-6: The interface shall allow for drag-and-drop functionality to customize the Performance Dashboard, enabling users to add, remove, and rearrange widgets such as embryo success rates and health metrics based on user preference.
UI-7: A dedicated settings page shall allow users to configure their alert preferences, such as enabling or disabling real-time notifications for embryo grading updates, pregnancy predictions, and health anomalies.
UI-8: The Ovulite Assistant module shall feature an interactive chat window that accepts text input from the user and provides AI-generated responses. The chat window shall display a scrollable history of the conversation and offer step-by-step guidance in real-time.

2.4.2 Software interfaces

SI-1: PostgreSQL 15+ – Used for persistent storage of the 39-column clinical dataset and user roles.
SI-2: FastAPI – Acts as the high-performance bridge between the React frontend and the Python ML models (XGBoost/PyTorch).
SI-3: PyTorch / TorchVision – Used for serving the **ResNet18** model for embryo grading. (Update this if you are using PyTorch instead of TensorFlow).
SI-4: XGBoost / TabPFN – Models for pregnancy prediction shall be loaded as serialized objects for real-time inference.

2.4.3 Hardware interfaces

HW-1: The system shall run on a standard VPS (e.g., AWS t3.medium) with **4GB to 8GB RAM**.
HW-2: CNN inference shall be performed on CPU in the initial deployment. GPU support is optional for future scaling.

2.4.4 Communications interfaces

CI-1: The Ovulite platform shall send an email or text message (based on user account settings) to veterinarians and farm managers to confirm the prediction of pregnancy outcomes and embryo transfer schedules.
CI-2: The system shall provide real-time web notifications for **Lab QC Anomaly Alerts**
CI-3: The Ovulite system shall allow veterinarians to send automated email reports with embryo grading results and pregnancy predictions to farm managers for further action and record-keeping.
CI-4: The system shall use HTTPS protocol to ensure secure data transmission.

CI-5: The platform shall provide a **Standardized Data Entry Form** to ensure all 39 required clinical variables are captured correctly before running a prediction.
CI-6: The system shall support integration with third-party messaging or notification platforms (e.g., SMS, email) to ensure communication with external systems and users who may not be logged into the Ovulite platform.
CI-7: Ovulite shall support automated system alerts via email or mobile notifications to administrators if any system errors or malfunctions occur, ensuring timely intervention.

Appendix -A

Plagiarism Report:

OVULITE_SCOPE_DOCUMENT.docx

ORIGINALITY REPORT

9%	7%	0%	8%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Higher Education Commission Pakistan Student Paper	3%
2	www.coursehero.com Internet Source	2%
3	Submitted to Shifa Tameer-e-Millat University (STMU) Student Paper	1%
4	Submitted to Lindenwood University Student Paper	1%
5	vehari.comsats.edu.pk Internet Source	<1%
6	Submitted to Stella Maris College Student Paper	<1%
7	Submitted to Ain Shams University Student Paper	<1%
8	Submitted to Myongji University Graduate School Student Paper	<1%
9	medinform.jmir.org Internet Source	<1%
10	open-innovation-projects.org Internet Source	<1%
11	www.lambdatest.com Internet Source	<1%
12	www.pl.vim.org Internet Source	<1%

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	On		